

Risk Management for Event-Driven Funds

Philippe Jorion

Paul Merage School of Business, University of California at Irvine, and
Pacific Alternative Asset Management Company (PAAMCO)

This version: September 2007

DRAFT

The author wishes to thank Jim Berens, Jane Buchan, Mayer Cherem, Rob Dudley, Judy Posnikoff, and Sydney Zhang for useful discussions on this topic.

Correspondence can be addressed to:

Philippe Jorion
Paul Merage School of Business
University of California at Irvine
Irvine, CA 92697-3125
Phone: (949) 824-5245
E-mail: pjorion@uci.edu

PAAMCO
19540 Jamboree Road
Irvine, CA 92612
Phone: (949) 261-4964
E-mail: pjorion@paamco.com

Risk Management for Event-Driven Funds

Abstract

Many portfolio strategies are “event-driven,” i.e., try to benefit from price movements caused by corporate events such as restructurings, bankruptcies, mergers, acquisitions, or other special situations. Such trading strategies involve payoffs that have discontinuous and skewed distributions that cannot be measured well with conventional risk methods. This paper develops methods to measure the forward-looking risk of portfolios exposed to such discrete events, based on current positions. When events are independent, the portfolio follows a binomial distribution. This approach is extended to the more realistic case where events are not independent. For mergers and acquisitions, empirical estimates of deal break correlations are positive but relatively low, which implies that most event risk is idiosyncratic and diversifiable. This methodology can be used to evaluate the risk and return of different portfolio structures.

JEL Classifications: G11 (portfolio choice), G23 (private financial institutions), G32 (financial risk management)

Keywords: merger arbitrage, risk management, hedge funds, value at risk

Introduction

Many hedge fund strategies are “event-driven,” i.e., try to benefit from price movements caused by corporate events such as restructurings, bankruptcies, mergers, acquisitions, or other special situations. Such trading strategies involve payoffs that have discontinuous distributions. Either the event happens or not, which is a binary distribution. These distributions pose a particular challenge to risk management because of their discontinuous nature, as well as the asymmetry of their payoffs. In addition, the history of price movements may not be relevant for measuring risk.

Consider, for example, a portfolio manager positioned to take advantage of a proposed merger and acquisition (M&A). The acquirer makes an offer that exceeds the current market price, by an amount known as the “merger premium.” Offers can take the form of cash or stock of the bidding company. Right after the deal announcement, the target company’s price jumps up, in proportion to this premium. There is still uncertainty, however, as to the success of the deal, which explains why the price of the target company generally trades at a discount to the offer price, known as the “arbitrage spread.” The typical position for risk arbitrageurs is to buy the target company and, in the case of stock deals, short the acquirer. If a deal succeeds, the position generally creates a modest gain.¹ In contrast, in the case of deal failure, the target company’s price can fall back to its pre-announcement level, which is a big loss.² Thus the position is exposed to an asymmetric “deal risk.” Such risk, however, cannot be measured from

¹ Success, however, does not simply mean that the original bidder takes control of the target. Other bidders can intervene in the meantime. Success can be defined as a situation where the price of the target goes to or above the offer price due to any acquisition. Equivalently, failure is the case when the target remains independent.

² The drop can be more than the initial run-up in price if the deal reveals negative information about the target. The payoff on the strategy also depends on the price movement in the acquirer’s stock in the case of positions that involve both the target and the acquirer.

historical data. The goal of this paper is to develop methods to measure the risk of portfolios exposed to such discrete events.

Current industry methods for managing the risk of such positions are rather elementary. Moore et al. (2006) describe a recent survey of risk management methods used by 21 risk arbitrageurs in mergers and acquisitions. The primary method used to control event risk is position limits. The first type is a limit on the fraction of the portfolio invested in any deal, which is typically 10% of the portfolio value. This implies a minimum of 10 positions. In the survey sample, the number of deals varies between 20 and 100, with a median of 30. The second type is a limit on the maximum loss of any position, e.g. 5% of the portfolio value if the deal fails. This loss represents a fall in the stock price back to its pre-announcement price. These two types of constraints are largely overlapping.

Such single name limits help to diversify the portfolio but are totally ad hoc. This is because, as revealed in the Moore survey, M&A arbitrageurs do not measure their *total* portfolio risk. Portfolio risk can be summarized, however, with a single measure of downside risk such as Value at Risk (VAR).³

The construction of portfolio risk measures, however, cannot be based on traditional risk models, which do not take into account the uncertainty generated by the currently unfolding events. Indeed, Schachter (2006) states that VAR “is not well suited” in these cases. This is certainly the case for conventional VAR measures based on recent historical data. Nevertheless, we show that it is perfectly feasible to construct a forward-

³ See Jorion (2006) for an exposition of VAR.

looking distribution of portfolio profits and losses based on current positions and to summarize it using a VAR measure.

Such measure would allow risk arbitrageurs to gauge optimum diversification levels, which depend on the number of deals, the expected profit on each deal, and the contribution of each deal to portfolio risk. Also, VAR can be used to determine the optimal amount of economic capital required to support the portfolio, or equivalently, the optimal amount of leverage. Moore et al. (2006) report that about two-thirds of arbitrageurs use leverage.

To our knowledge, this paper is the first to provide quantitative tools to measure the portfolio risk of event-driven funds. The methodology proposed here expands on methods developed for credit portfolios, which also involve discrete events, i.e., defaults. In the case of cash deals, a position in the target firm is similar to an unsecured bond issued by the acquirer that matures on the closing date. The methodology further demonstrates the benefits of using position information to manage risks.

The paper proceeds in two steps. Section I describes drawbacks of conventional risk measures for event-driven funds. Section II then derives a new methodology for measuring portfolio risk when events are independent. Realistically, however, there may be dependencies across events. Section III provides some empirical evidence on probabilities and correlations of deal breaks. Section IV then analyzes portfolio risk with correlated events and shows how correlations affect risk measures for event-driven portfolios. Section V draws implications for assessing the amount of economic capital required to support the portfolio as well as acceptable levels of leverage. Finally, Section VI contains some conclusions.

I. Drawbacks of Conventional Risk Measures

Typical risk measures are returns-based. One example is the volatility of historical returns for the fund over a recent window. This approach is simple but has severe shortcomings. It is unable to react quickly to changes in the portfolio composition or in the trading strategy. In addition, it makes it difficult to understand the structural drivers of risk and hence to manage the risk profile of the portfolio.

More recently, position-based risk measures such as VAR have become widely used. Conventional VAR measures combine current position information with the recent history of risk factors, assuming that this history can be applied to the future. Current positions are taken as fixed over the risk horizon.

This approach could be defined as “conventional position-based.” Generally, this is a vast improvement over returns-based methods because it can handle changes in portfolio composition and can be used to manage portfolio risk.

In the case of event-driven funds, however, the conventional application of position-based VAR methods can be misleading for a number of reasons. First, the event itself changes the nature of the stochastic process, from a random walk to a discrete-outcome process. Thus, historical data cannot be used in the usual fashion. As an example, consider the data provided by Mitchell and Pulvino (2001). They report that, over the period 1963 to 1998, a value-weighted portfolio of M&A deals had an annual realized volatility of 9.3%. This number represents the actual risk of a portfolio invested in an average of 31 M&A positions taken after the announcement. They do not provide the volatility of the same portfolio before the announcement, but surely this should be less than that of a value-weighted market index because the latter is better diversified.

Over the same period, this index had a volatility of 15.1%. This number can be taken as a lower limit for the position-based risk of the M&A portfolio. In this case, the actual risk is $(15.1-9.3)/15.1=38\%$ lower than the conventional position-based risk, which is substantial.

Another problem is that the recent history includes the announcement of the event itself. This creates a large jump in the price of the target, which artificially increases the risk measure. The shorter is the window, the larger is the effect. As an example, take an unleveraged M&A hedge fund with more than 150 positions. From May 2005 to April 2007, this fund had a return-based volatility of 2.2% pa. Conventional risk forecasts were also measured for this fund due to the availability of position data, applied each month to a 4-year window. Over the same period, these monthly volatility forecasts averaged to 9.8%. Here again, the bias is substantial. Actual risk is about $(9.8-2.2)/9.8=78\%$ less than the conventional position-based risk measure.

Perhaps simple rules of thumb could be developed for correcting conventional position-based risk measures. Such adjustments cannot be universal, however. This is because the relationship between a conventional position-based risk measure and its actual counterpart depends on the structure of the portfolio, which involves the number of positions, the distribution of deal payoffs, as well as historical risk measures. Note, for instance, that the actual risk of the first M&A portfolio was 9.3%, which is much higher than the risk of the second M&A portfolio, which was 2.2%. This is because the first portfolio was much less diversified, with 31 positions against 150 for the second.

Instead, position-based information can be used directly to construct forward-looking measures of risk that account for the discrete-outcome processes that characterize event-driven funds. This is the main purpose of this paper.

II. Measuring Risk from Position Information

Suppose we are investing in a stock that is currently in play as a takeover target at a fixed offer price. The distribution of payoffs is characterized by two states of the world. Table 1 gives an example. We evaluate a cash deal where the stock of the target currently trades at \$100, post-announcement. We ignore directional market risk and focus on event risk only. The event indicator is b . In the case of failure or deal break, $b=1$, with probability $p=0.15$; the dollar return R is an absolute loss AL of $-\$15$. In case of success, $b=0$, the absolute profit is $AP=+\$5$.

Table 1: Single-Deal Distribution

	Failure ($b=1$)	Success ($b=0$)
Probability	15%	85%
Payoff (R)	$-\$15$	$+\$5$

This example assumes a probability of failure of $E[b] = p = 15\%$, which is typical of empirical studies of M&A.⁴ More generally, the portfolio manager is supposed to assess the probabilities and payoffs of each deal and to add value through skillful selection of deals. In this example, the expected profit is positive, at $E[R] = 15\% \times (-\$15) + 85\% \times (\$5) = \$2.00$. For simplicity, this is assumed to be over the risk-free rate. The

⁴ See for instance Baker and Savasoglu (2002).

volatility for this Bernoulli distribution is $\sigma[b] = \sqrt{p(1-p)}$. The distribution, however, is strongly skewed in the direction of losses.

Here, the horizon is assumed to be the average time needed for the resolution of a deal. Mitchell and Pulvino (2001) indicate that the average horizon of a deal is 59 trading days, or about three calendar months. In practice, this horizon is uncertain. The distribution of payoffs could be extended to include an intermediate case where the deal is still pending. Extrapolating the quarterly horizon to a year, the expected rate of return is $(\$2 \times 4)/\$100 = 8$ percent per annum.⁵

Note that this example is calibrated to be realistic in terms of expected returns: Mitchell and Pulvino (2001) find that simulated excess returns from risk arbitrage strategies average 9.25% per annum, which is equivalent to the 2% per quarter assumed here.⁶ These numbers represent pure alpha because they adjust for the risk-free rate and the market beta. The methodology presented here, however, does not depend on these numbers.

The next issue is the aggregation at the portfolio level. This type of problem is similar to building a distribution of portfolio credit losses. To simplify, define L as a positive number representing the loss relative to the success case, e.g., \$20 for Table 1. Assume first a homogeneous portfolio: All deals have equal size L and equal failure probability p . If the events are independent, it is straightforward to build this distribution.

⁵ A variety of explanations have been advanced for this “risk arbitrage” premium. One is that risk arbitrageurs need compensation for bearing event risk. This information-rich environment requires specialized expertise, e.g. trading and legal, which needs compensation. Another is that some investors may be forced to sell because of investment guidelines that would be breached after a takeover. For instance, a minimum level of portfolio yield could be breached if the merged company pays no dividend. Finally, these strategies have non-linear exposures to the market, because the break probability increases when the market falls. This is akin to a short put position, which should command a risk premium.

⁶ After costs, however, they report an average return of about 3.5%.

The total number of deals is N . The total number of failures k then follows the binomial distribution f :

$$f(k; N, p) = \binom{N}{k} p^k (1 - p)^{N - k} \quad (1)$$

The distribution of total portfolio losses $TL = k \times L$ is given directly by Equation (1). This is also a binomial distribution, which can be summarized by a dispersion measure, such as VAR, or the lowest quantile at a specified confidence level.

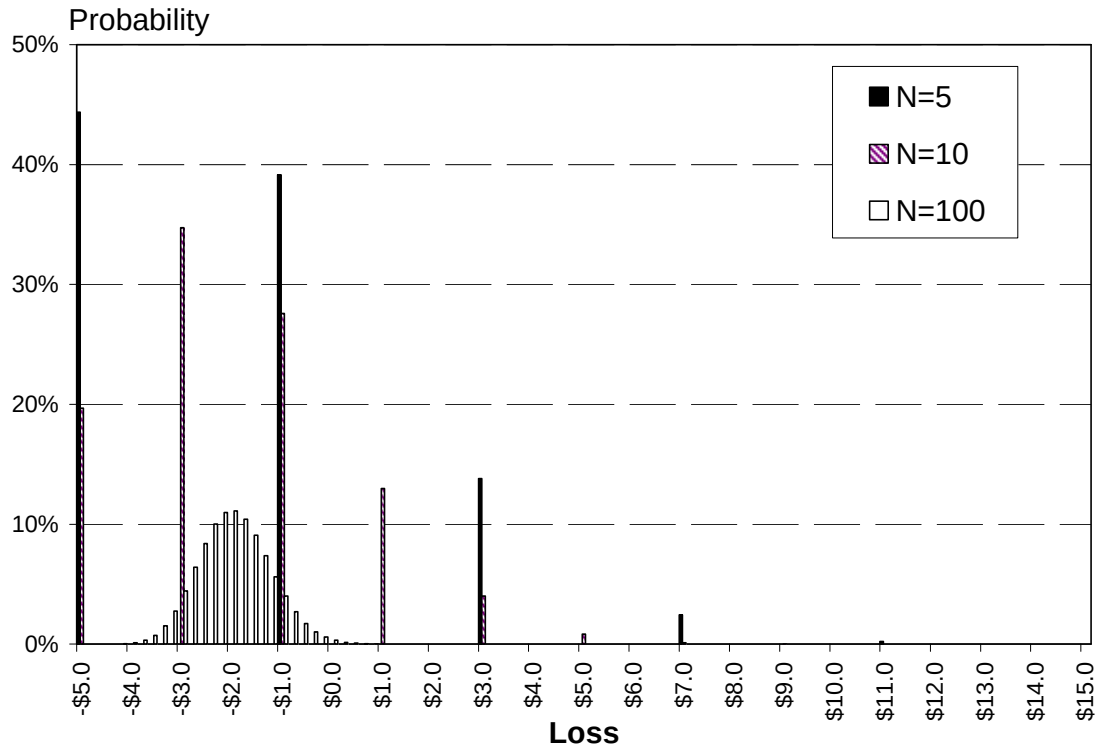
We can now explore the effect of changing the number of deals N on the shape of the distribution. To maintain comparability across portfolios, the total dollar exposure is kept fixed, say at \$100. Hence, the size of each deal L is inversely related to N across portfolios. We characterize the distribution by its VAR, say at the 95 percent confidence level. Because the distribution is discrete, we report the loss associated with a confidence level at least equal to 95 percent. Table 2 reports the VAR along with the associated actual confidence level and number of failures. VAR is measured as the negative of the dollar loss over the period, and is generally a positive number. The distribution of losses is described in Figure 1 for $N=5, 10$, and 100 .

Table 2: Distribution of Independent Deals (Constant Exposure of \$100)

Number of Deals	Expected Profit	VAR($\geq 95\%$)			VAR($\geq 99.9\%$)		
		VAR	Actual Confidence Level	Number of Failures	VAR	Actual Confidence Level	Number of Failures
1	+\$2.00	\$15.00	100.0%	1	\$15.00	100.0%	1
5	+\$2.00	\$3.00	97.3%	2	\$11.00	99.99%	4
10	+\$2.00	\$1.00	95.0%	3	\$7.00	99.99%	6
20	+\$2.00	\$1.00	97.8%	6	\$4.00	99.98%	9
30	+\$2.00	\$0.33	97.2%	8	\$2.33	99.92%	11
40	+\$2.00	\$0.00	97.1%	10	\$2.00	99.96%	14
50	+\$2.00	-\$0.20	97.0%	12	\$1.40	99.93%	16
100	+\$2.00	-\$0.80	96.1%	21	\$0.40	99.94%	27

Table 2 shows that the expected dollar profit is independent of the number of deals, which simply reflects the fixed total exposure of the portfolio. VAR, however, does change with the number of deals. With one deal only, the minimum 95% VAR implies a loss of \$15, which actually corresponds to a 100 percent confidence level, or one failure. With 10 independent deals, the 95% VAR goes down to \$1.00. With 40 deals, the 95% VAR is zero. In other words, there is a low probability (less than 5%) of losing any money on this portfolio. This is because the expected profit is positive and the dispersion of payoffs shrinks rapidly as N increases. As the number of deals further increases to 100, VAR becomes $-\$0.80$, which means that there is a low probability of making a profit less than \$0.80. Thus, increasing the number of independent deals shrinks the spread of the distribution rather rapidly, as illustrated in Figure 1. As N increases, the binomial distribution converges to a Poisson distribution.

It should also be noted that, because of the discrete nature of outcomes, VAR does not always change smoothly when changing parameters. Consider for instance VAR at the minimum 95 percent confidence level. This stays at \$1.00 for $N=10$ and 20. For $N=10$, the actual confidence level is just at its required minimum of 95.0%. At $N=20$, the actual confidence level increases to 97.8%. This corresponds to 6 deal failures; the confidence level for 5 deal failures is still below 95%. As N increases further, VAR eventually drops, but in discrete steps. Thus VAR is not a smooth function.

Fig. 1. Distribution of Portfolio Losses with Independence

This distribution could be used to infer the maximum amount of leverage that the fund could safely take. Assuming the fund wants to maintain a single-A credit rating, the confidence level can be taken as one minus the default probability over a year. Credit rating agencies report a historical default rate of approximately 0.1 percent for A-rated companies. This implies a confidence level of 99.9 percent.⁷ This VAR measure, which can be interpreted as “economic capital,” is listed in the last column of Table 2.

For one deal, the 99.9% VAR is \$15. If the notional value of the position is \$100, the portfolio can support a maximum leverage of \$100/\$15, or 6.7. For 10 deals, economic capital shrinks to \$7.0. Thus leverage could go to \$100/\$7.0, or 14.3. This

⁷ Under the newly established Basel II rules, commercial banks need to maintain regulatory capital in excess of their 99.9 percent credit VAR over a horizon of one year. In our case, the horizon is one quarter, so that the annual default rate is even lower than that for A-rated credits.

would generate a rate of return on economic capital of \$2.0/\$7.0, or 28.6% per quarter.⁸

With 100 deals, economic capital is \$0.4. Because this number is so low, leverage can be even higher, generating a rate of return of 500%. Of course, this very high number cannot be realistic because it assumes independent deals. The effect of dependencies is examined next.

⁸ Normally, financing costs should be taken into account. The estimate of 2% expected return, however, is already in excess of the risk-free rate, so there is no need for further adjustment. The only portion of financing costs that is ignored is the spread between the borrowing and deposit rate.

III. Empirical Estimation of Event Correlations

In practice, the events in the portfolio may not be independent. A large fall in the market could lead to many deals breaking during the same year. Mitchell and Pulvino (2001), for example, show that a 20 percent decrease in the market return increases the probability of failure by about 9 percent. This contemporaneous dependency on the market is akin to positive correlations across events.

This section provides empirical estimates of the deal-break probability and the average break correlation. This study considers a large sample of 1,765 M&A deals involving at least one North American company and concluded between 1997 and 2006. This sample only includes deals with an announced total value over \$500 million. The data were collected from Bloomberg and are summarized in Table 3. Deals are tabulated by their outcome every quarter.

Table 3: Description of M&A Deals: 1997-2006

	Number of Deals Completed per Quarter	Average Value of Deals (\$ million)	Mean Completion Days for Deals
Failure	5.3	\$5,377	138
Success	38.8	\$3,385	137
Total	44.1		

In this 40-quarter period, we observe 212 deal breaks and 1,553 completed deals, which gives a 12% failure rate. The average number of deals completed in a quarter is 44.1, with a mean completion time of 138 calendar days, or about 4.5 months.

To estimate correlations across deal failures, we use a standard methodology developed for credit events.⁹ Initially, each deal is assumed to have the same

⁹ See for example Bahar and Regal (2001) and de Servigny and Renault (2002).

unconditional probability of failure p , and all pairwise correlation coefficients ρ are assumed equal. This is extended later.

Suppose that we observe a number F_t of failures during a period t , out of a total of N_t observations. The probability of failure is then estimated by

$$p = E[b] = \sum_{t=1}^T w_t \frac{F_t}{N_t} \quad (2)$$

where w_t is the weight assigned to each observation, here taken as $(1/T)$. The probability p_{12} of a joint deal break can be estimated from the ratio of the total number of pairs breaking during the period, or $F_t(F_t-1)/2$, to the total number of pairs combinations, or $N_t(N_t-1)/2$. De Servigny and Renault (2002) show via simulations that small-sample properties of the joint default probability estimator are slightly better for

$$p_{12} = E[b_1 b_2] = \sum_{t=1}^T w_t \frac{(F_t)^2}{(N_t)^2} \quad (3)$$

The deal break correlation can then be constructed from the covariance between the two events

$$\text{Cov}(b_1 b_2) = \rho \times \sigma(b) \sigma(b) = E[b_1 - p][b_2 - p] = E[b_1 b_2] - p^2 = p_{12} - p^2 \quad (4)$$

Replacing $\sigma(b)$ by the standard deviation of a Bernoulli variable, the correlation can be written as

$$\rho = \frac{p_{12} - p \times p}{\sqrt{p(1-p)} \times \sqrt{p(1-p)}} \quad (5)$$

Note that the deal break correlation is also related to the variance of the time series of fraction of breaks during each period. Define this fraction as $x_t = F_t/N_t$. If x_t fluctuates sharply over time, this implies clustering of failures, or positive correlation. Using

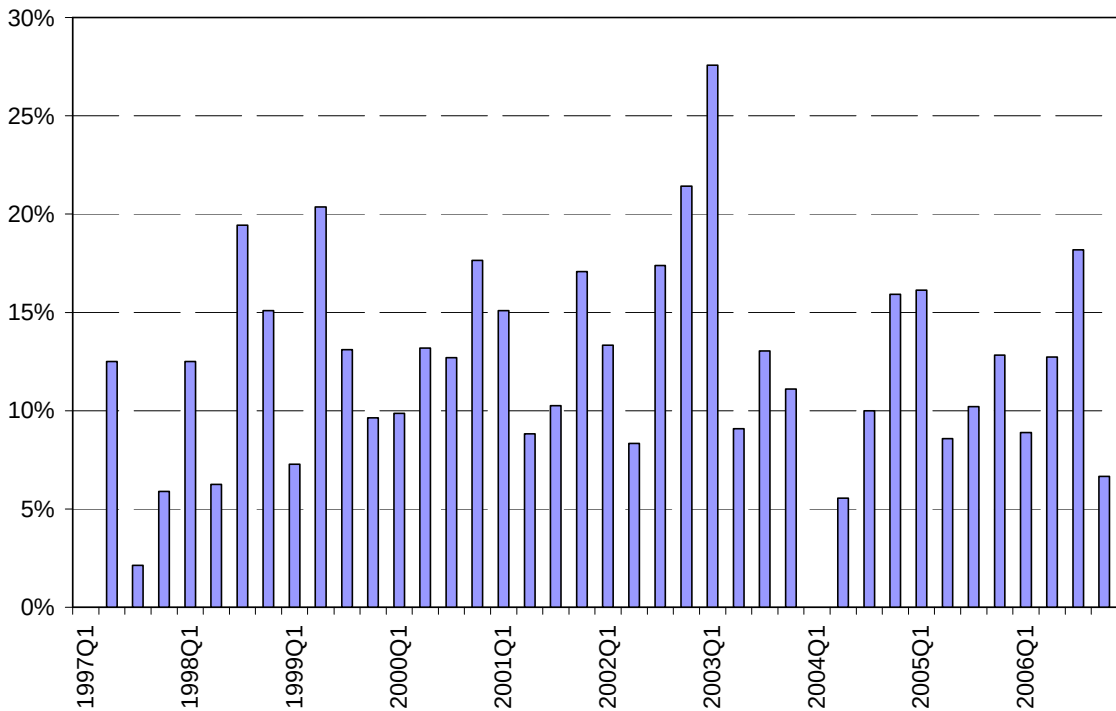
Equations (2), (3) and (4), and ignoring the degree of freedom adjustment, the variance of x can be written as

$$\sigma^2 = \frac{1}{T} \sum_{t=1}^T (x_t - \bar{x})^2 = \left(\frac{1}{T} \sum_{t=1}^T x_t^2 \right) - \bar{x}^2 = p_{12} - p^2 = \rho \times p(1-p) \quad (6)$$

So, a high variance is equivalent to high failure correlation.

Figure 2 plots the quarterly fraction of deals failing. Using Equation (2), the average break probability is estimated to be 12%. The figure, however, reveals some variation in the fraction of deals failing over time. This reached a peak of 27% in the first quarter of 2003. The fraction was zero for two quarters (the first quarters of 1997 and 2004.) This variation can be interpreted in terms of clustering of failures, or correlations across deal breaks.

Fig. 2. Fraction of Deals Failing: 1997-2006, Quarterly



Using Equation (3) to estimate p_{12} and Equation (5), the break correlation is estimated at 0.03. The failure probability is 0.12. To assess the reliability of these estimates, we simulate their distribution by bootstrapping the sample of 40 quarterly fractions. Based on 1000 replications, the 95% significance interval for the break probability is (0.102, 0.137), and that for the break correlation is (0.017, 0.045). Therefore, the break correlation is positive and significantly different from zero. This justifies the use of an average break correlation of 0.03 in the following section.

More generally, the portfolio manager could model the probability of failure as a function of a number of deal variables.¹⁰ Here, we fit the aggregate fraction of breaks using stock market returns and bond yields as independent variables. We can distinguish between “predictive regressions,” which model predictable time-variation in the failure probability, and “explanatory regressions,” which model correlations through the influence of common risk factors. Results are presented in Table 4.

The first column in Table 4 shows a negative coefficient on the stock market return during the same quarter. This means that the fraction of breaks increases when the stock market falls during the same quarter. Splitting up market returns into positive and negative values shows that the effect is asymmetric. Positive returns are basically uncorrelated with deal breaks. Large negative returns, however, are significantly associated with a greater proportion of deal breaks. This is as expected: A large drop in the stock market reduces the value of the target and makes it more likely for the acquirer to walk away from the deal.

¹⁰ Mitchell and Pulvino (2001) estimate the probability of a deal breaking from individual deal characteristics and general market factors. For example, “hostility” is the most important indicator, increasing the probability of failure by 12.8 percent. They also find that the S&P beta on the downside is greater than on the upside.

The -0.36 coefficient means that when the stock market drops by 20%, the break probability increases by 7%. This conditional dependence on the market explains why we observe positive correlations across deal breaks.¹¹ After conditioning on this common factor, the break correlation becomes unimportant. In practice, using a simple measure of unconditional positive correlation is sufficient because market downturns cannot be predicted.

Table 4: Explaining and Predicting Fraction of Breaks for M&A Deals

This table presents regressions of the fraction of breaks on various contemporaneous and predictive variables. RM is the total return on the S&P500 index; HY is the yield to maturity on the Lehman high-yield corporate bond index. Sample period is 1997-2006, or 40 quarters. Standard errors are below the estimated coefficients. The superscripts **, and * indicate significance at two-tailed 5% and 10% levels, respectively.

Independent Variables	Explanatory Model 1	Explanatory Model 2	Predictive Model 1	Predictive Model 2	Predictive Model 3
Constant	0.12 (0.01)	0.10 (0.01)	0.13 (0.01)	0.02 (0.04)	0.05 (0.04)
RM(t)	-0.08 (0.10)				
RM(t)>0		0.13 (0.18)			
RM(t)<0		-0.36* (0.22)			
RM(t-1)			-0.26** (0.10)		-0.19* (0.10)
HY(t-1)				1.00** (0.40)	0.68 (0.42)
R-square (%)	1.67	7.14	14.30	15.54	21.04

Focusing next on predictive regressions, the coefficient on the previous stock market return is negative and strongly significant, indicating that falling stock prices increase the fraction of breaks the next quarter. The explanation is the same as before. In addition, the coefficient on the bond yield at the start of the quarter is positive and highly significant. As expected, higher financing costs lead to a greater fraction of deals failing.

¹¹ Note that a similar factor structure is used for credit portfolio models, where correlations across defaults are induced by a 1-factor structure.

This leads to a predictable break probability that varies between 9% and 18% over this period.

It should be noted that this calibration exercise is limited to large North American deals. To lower the failure correlation, a portfolio manager could include in the portfolio M&A deals from smaller firms and from outside North America. More generally, the methodology described next can be used to illustrate how to assess the benefits from searching for lower correlations.

IV. Building Portfolio Distributions with Correlated Events

Section I has shown that the dispersion of the distribution of losses shrinks rapidly as the number of independent deals increases. In practice, however, positive correlations temper this risk reduction effect. The question is how to adjust the measure of portfolio risk. Because M&A events are similar in structure to credit events, the methodology behind recent credit risk models can be used to measure event risk.

Various methods have been proposed to measure the risk of credit portfolios and can be applied to this case. One approach is the “Binomial Expansion Technique” (BET) advocated by Moody’s (1996) to evaluate Collateralized Debt Obligations (CDOs). This technique approximates the actual distribution of N correlated events with a simpler binomial distribution for another number D of uncorrelated events. D is called the “diversity” score. It is computed for each deal i , from event probabilities (p) deal size (L), and bivariate correlations (ρ):

$$D = \frac{\left(\sum_{i=1}^N p_i L_i \right) \left(\sum_{i=1}^N (1 - p_i) L_i \right)}{\sum_{i=1}^N \sum_{j=1}^N \rho_{ij} \sqrt{p_i (1 - p_i) p_j (1 - p_j)} L_i L_j} \quad (7)$$

This expression is obtained by matching the first and second moments of the true distribution with the binomial distribution $f(k; D, p)$. This is therefore an approximation only. As before, VAR can be computed for this new distribution.

To illustrate, assume that all probabilities, all losses, and all correlations are equal, even though this need not be the case. The diversity score then reduces to

$$D = \frac{(N p L)(N (1 - p) L)}{(N + \rho(N - 1)N) \sqrt{p(1 - p)p(1 - p)} L L} = \frac{N^2}{(N + \rho(N - 1)N)} \quad (8)$$

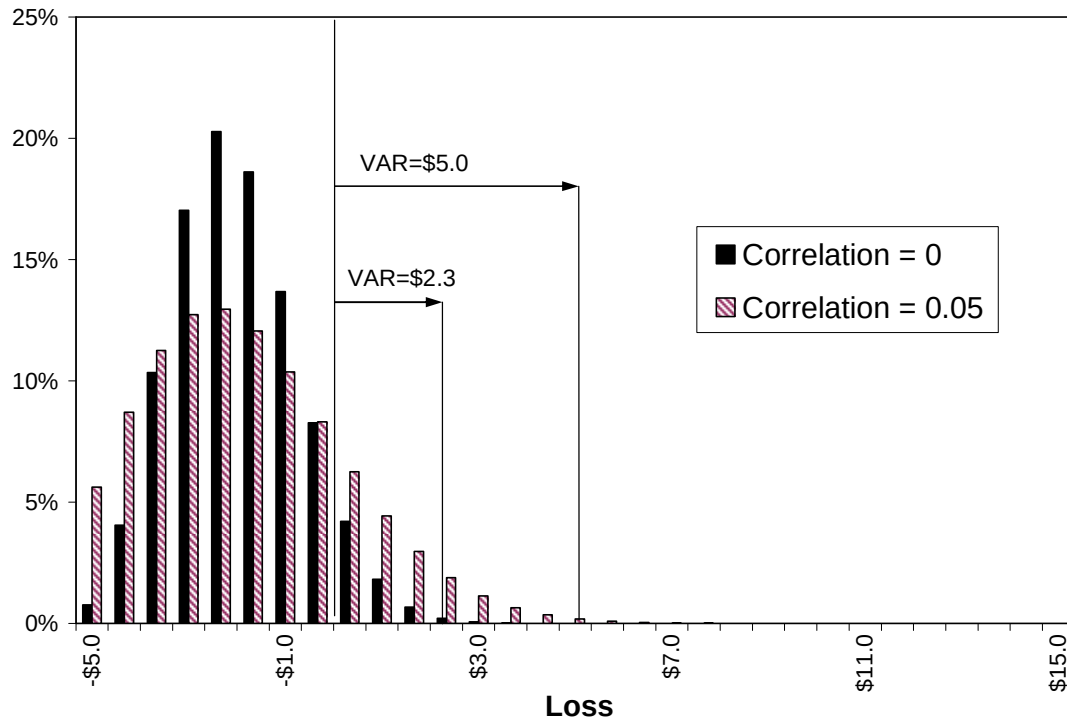
If the events are independent, all correlations are zero, and Equation (7) reduces to $D=N$, as we would expect. More generally, a positive correlation leads to $D<N$, which implies that the distribution has longer tails than with independent events.

Table 5 compares the distributions of portfolio of $N=30$ deals for different correlation coefficients. The zero correlation case is the same as before, with a 95% VAR of \$0.33. The diversity score decreases with the correlation. For example, when the average correlation increases to 0.03, the diversity score falls from 30 to 16. Accordingly, the 95% VAR increases sharply, from \$0.33 to \$1.25. With a correlation of 0.10, the 95% VAR moves to \$3.57. As expected, higher correlations lead to greater risk. When the correlation is 0.50 or above, VAR moves to \$15, which is the same as for a portfolio of one deal. In this case, there is no diversification benefit.

Table 5: Distribution of Dependent Deals Using the BET Approach (N=30)

Correlation	Diversity Score	Expected Profit	VAR ($\geq 95\%$)	VAR ($\geq 99.9\%$) [Actual Confidence Level]	
0.00	30.0	+\$2.00	\$0.33	\$2.33	[99.92%]
0.03	16.0	+\$2.00	\$1.25	\$5.00	[99.98%]
0.05	12.2	+\$2.00	\$1.67	\$5.00	[99.93%]
0.10	7.7	+\$2.00	\$3.57	\$9.29	[99.99%]
0.20	4.4	+\$2.00	\$5.00	\$10.00	[99.95%]
0.50	1.9	+\$2.00	\$15.00	\$15.00	[100.00%]

Figure 3 compares the distribution of losses with $N=30$ in two cases, $\rho=0$ and $\rho=0.05$. The distributions are heavily skewed. In each case the expected return is \$2.0. With zero correlation, the right tail falls off quickly. With positive correlation, the tail is much longer. Thus, a modest amount of correlation has a substantial impact on portfolio risk.

Fig. 3. Distribution of Portfolio Losses with Various Correlations: N=30

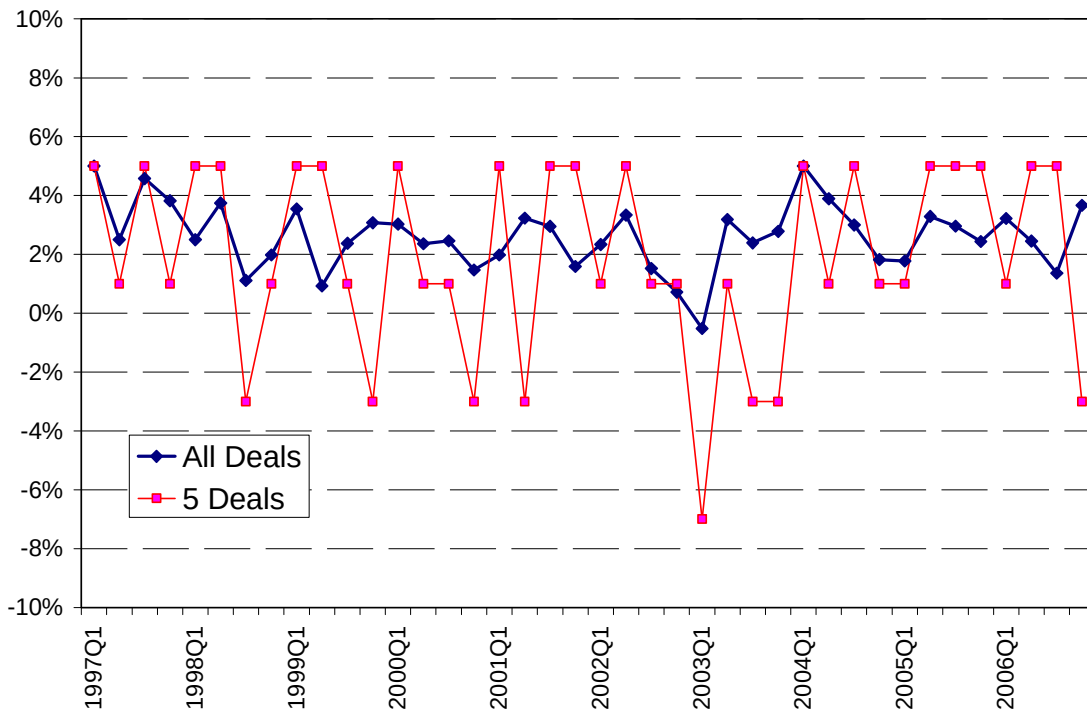
More general methods can be used instead of the Binomial Expansion Technique approach. The distribution of events can be constructed from Monte Carlo simulations. In this framework, each event is driven by a latent random variable such that a drop below some cutoff point simulates a break with a specified probability. Correlations can be induced across these random variables so as to generate specified default correlations.¹² Such bottom-up approach, which is similar to structural models of credit risk, can be used to derive a more general distribution of gains and losses for the event-driven portfolio. This will be used for results in the next section.

As an illustration of the effect of number of deals on the risk profile of the portfolio, a simple experiment can be performed based on the actual number of deals

¹² In our example with $p=15\%$, we need a correlation of latent variables calibrated at 0.068 to obtain the observed default correlation of 0.03 using multivariate normal random variables.

breaks and successes over the period 1997 to 2006. The average number of deals per quarter is 44. The first portfolio is equally-invested in all deals. Consistently with the previous analysis, a success is assigned a return of +5%; a failure has a return of −15%. Figure 4 displays the hypothetical time series of returns on this portfolio. During the first quarter, for example, the failure rate is exactly 0%. The return is then computed as $R = 0\%(-15\%) + (1-0\%)(+5\%) = 5\%$, which is the highest feasible value. Over the entire sample, the average return is close to 2%. The all-deal portfolio has very low volatility. In this experiment, it loses money in one quarter out of 40 only.

Fig. 4. Time-Series of Portfolio Returns for Different Portfolio Sizes



The second portfolio consists of five deals only. Each quarter, the outcome is selected randomly, using the deal break fraction in that same quarter as probability of failure. This approach preserves the pattern of correlations in the data. Figure 4 shows a

typical outcome. This 5-deal portfolio is much more volatile than the all-deal one even though it has the same mean. It suffers losses during eight quarters. The worst loss is -7% . Clearly, the portfolio with a greater number of deals is the safer one.

Alternatively, this first portfolio could be leveraged three times and have the same level of risk as the 5-deal portfolio, while delivering three times the expected return.

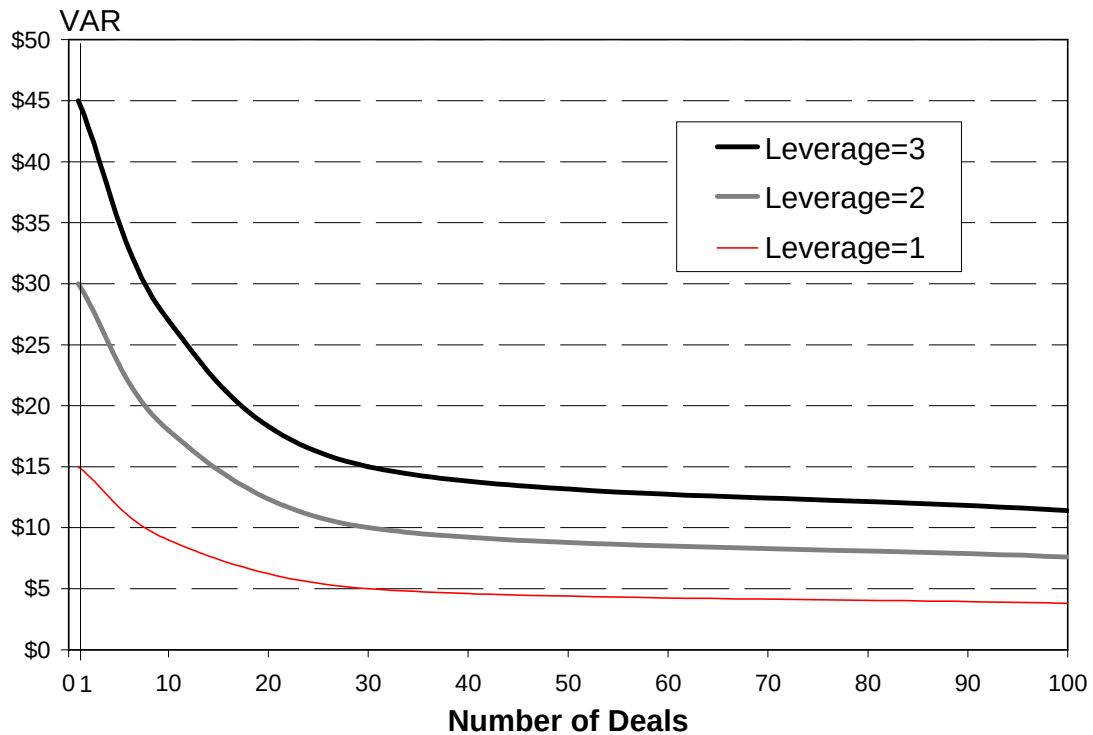
V. Implications for Assessing Economic Capital and Leverage

This methodology can be used to infer economic capital levels, or conversely the number of deals required to keep a fixed level of risk for event-driven portfolios.

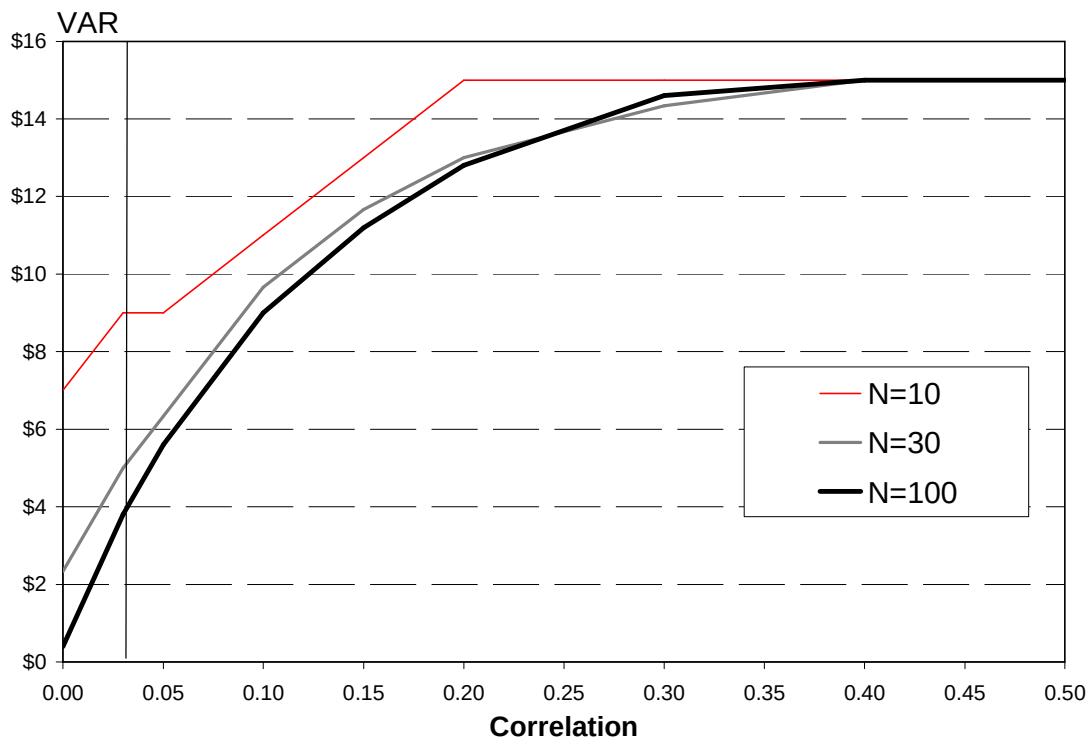
Leverage can be assessed from economic capital. Recall that with one deal only, economic capital is \$15, implying a maximum leverage of $\$100/\$15.0=6.7$ at which point the rate of return on economic capital is $\$2.0/\$15.0=13.3\%$ per quarter.

For a 10-deal portfolio with a correlation of 0.03, VAR is \$9.0, at a minimum 99.9% confidence level. As a result, the portfolio could be leveraged by a factor of $\$100/\9.0 , or about 10 times the original equity investment, generating an expected return of $\$2.0/\$9.0=22\%$.

This analysis shows that increasing the number of deals substantially lowers risk, because deal risk is largely idiosyncratic. When N increases to 30 and 100, economic capital drops to \$5.00 and \$3.80, respectively. Figure 5 illustrates how economic capital decreases with the number of deals. Lower risk allows greater leverage, increasing expected returns.

Fig. 5. Economic Capital and Portfolio Size (Break Correlation=0.03)

This powerful diversification effect is driven by the low correlation, taken as 0.03 here. Next, Figure 6 examines how correlations affect economic capital, for various portfolio sizes. The effect is rather dramatic. For $N=100$, increasing the correlation from 0.00 to 0.10 pushes economic capital from \$0.4 to \$9.0. Beyond a correlation of 0.30, there is basically no diversification effect: VAR converges to the worst loss of \$15. Average correlations are very unlikely to reach such values, however, from historical experience. The previously computed 95% confidence interval for ρ ranges from 0.017 to 0.045. This translates into a confidence interval of \$2.3 to \$5.2 for VAR with $N=100$. So, even in a worst-case scenario for the break correlation, there is still a fair amount of diversification.

Fig. 6. Economic Capital and Break Correlation (Break Probability=15%)

Economic capital is also sensitive to the break probability, but less so. Figure 7 shows that if this changes from the base number of 15% to 20%, economic capital increases from \$3.8 to \$5.0%, when $N=100$. The 95% confidence interval for the estimated break correlation was (10.2%, 13.7%) for this sample. So, the worst increase from the mid-point probability is only 2%. On the other hand, increases in the break probability are important because they directly reduce the expected return on the portfolio.

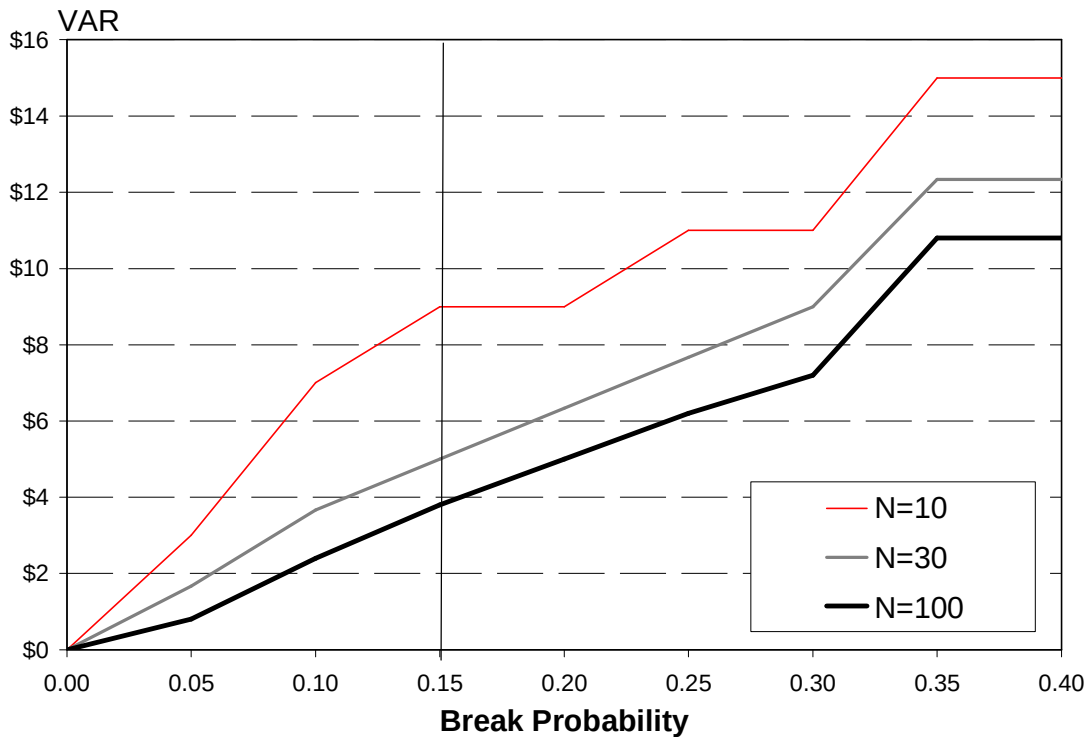
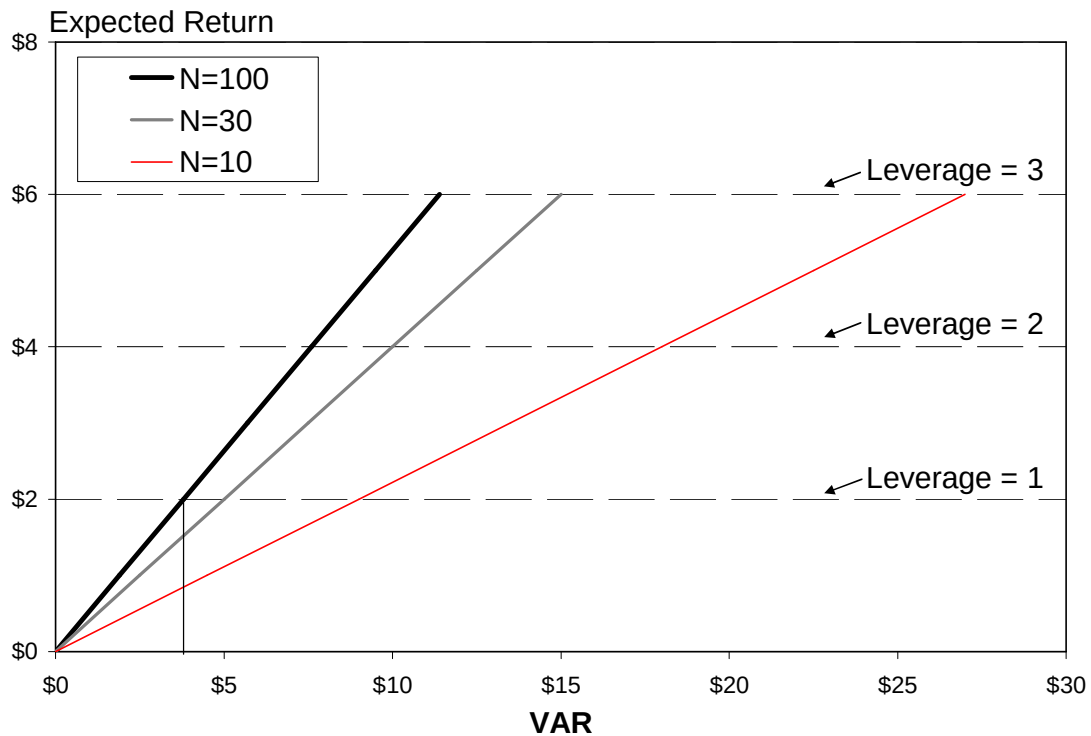
Fig. 7. Economic Capital and Break Probability (Break Correlation=0.03)

Figure 8 illustrates the trade-off between expected return and risk for various levels of leverage and portfolio size, assuming a baseline correlation of 0.03. Different levels of leverage are represented by horizontal lines. For a given portfolio size N , expected (excess) returns and risk increase linearly with leverage. A portfolio with $N=100$ can be 3 times leveraged and still have a level of risk similar to a portfolio with $N=10$, while delivering three times the expected return.

Apparently, this analysis would lead to the conclusion that having a larger number of deals is always an improvement. This is because we assumed that the expected profit per deal remains constant at \$2. In practice, portfolio managers have expertise in evaluating some types of deals. When we increase the number of deals to evaluate, we would expect that the profit per deal should decrease, reflecting acquisition information costs. Thus, there should be an optimal number of deals, reflecting the tradeoff between

risk and profits. Without a measure of portfolio risk, however, such assessment is not even possible.

**Fig. 8. Trade-Off between Expected Return and Economic Capital
(Break Correlation=0.03)**



Finally, we should note that the analysis of economic capital should also account for non-event risks, including market risk and some idiosyncratic risk.¹³ These additional risks can be incorporated in an enlarged model. Stress tests should also be performed to evaluate the sensitivity of the results to changes in input parameters, including break probabilities and correlations.

¹³ There is also horizon risk, which is the risk of a lengthening of the horizon for the deal resolution. This happened, for instance, during 2004, when the Sarbanes-Oxley Act temporarily slowed down deal activity.

VI. Conclusions

Event-driven portfolio strategies face discrete deal risk. In contrast with other types of market risks, where distributions are continuous, deal risk is represented by a binary variable, which is the success or failure of a deal. This type of risk cannot be captured by conventional position-based risk measures.

Current positions, however, can be used to construct the distribution of portfolio returns. To do so, the risk manager needs estimates of the probability of success for each deal, of the payoffs from success and failure, and of joint correlations across deals. This paper develops new methods to construct quantitative measures of portfolio risk for event-driven portfolios.

Once constructed, this distribution can be summarized by the usual risk measures, such as Value at Risk, which can be used to assess acceptable levels of leverage. This paper finds that, in the case of independent deals, the economic capital required to support a portfolio of 30 independent deals is six times lower than for a single-deal portfolio. Thus, this diversified portfolio could be levered six times more than a single-deal position and still maintain the same level of risk. In the meantime, this higher leverage implies a higher return on capital.

In practice, however, these diversification benefits are limited because of clustering across deal breaks. This paper provides empirical estimates indicating that the average deal-break correlation is around 0.03 for large North American deals. This positive correlation decreases the benefits from diversification and increases the required economic capital.

This methodology can be used to improve current risk management practices for event-driven portfolios. Apparently, portfolio managers primarily rely on position limits to control risk. Similar methods were used to manage credit risk before the generalized application of modern risk measurement methods.

Without such quantitative methods, it seems difficult to assess the optimal risk-return profile of the portfolio. With limited expertise or resources, increasing the number of deals is costly, because it decreases the average expected return per deal. On the other hand, increasing the number of deals does provide diversification benefits. The optimal portfolio represents the point where the marginal cost equates the marginal benefit from increasing deals. Without a formal measure of the marginal benefit, however, this optimal position cannot be identified.

Armed with these portfolio measurement tools, the portfolio manager should search for deals that are profitable, yet not too correlated across each other. This paper provides the means to assess the costs and benefits of different structures for event-driven portfolios.

REFERENCES

- Bahar, Reza and Krishnan Regal, 2001, “Measuring Default Correlation,” *Risk* (March), 129—132.
- Baker, Malcolm and Serkan Savasoglu, 2002, “Limited Arbitrage in Mergers and Acquisitions,” *Journal of Financial Economics* 64, 91-115.
- De Servigny, Arnaud and Olivier Renault, 2002, “Default Correlation: Empirical Evidence,” Working Paper, Standard and Poor’s.
- Jorion, Philippe, 2006, “Value at Risk: The New Benchmark to Manage Financial Risk,” McGrawHill.
- Mitchell, Mark and Todd Pulvino, 2001, “Characteristics of Risk and Return in Risk Arbitrage,” *Journal of Finance* 56, 2135-2175.
- Mitchell, Mark, Todd Pulvino, and Erik Stafford, 2004, “Price Pressure around Mergers,” *Journal of Finance* 59, 31-63.
- Moody’s, 1996, “The Binomial Expansion Technique Applied to CBO/CLO Analysis,” Moody’s Special Report.
- Moore, Keith, Gene Lai, and Henry Oppenheimer, 2006, “The Behavior of Risk Arbitrageurs in Mergers and Acquisitions,” *Journal of Alternative Investments* (Summer), 19-29.
- Schachter, Barry, 2006, “Limits on far VAR,” *Risk* (March).